Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/09/2005
PI Name:	Doerr, Harold K. M.D.		
Project Title:	MOST Proposal		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	11/01/2004	**End Date:**	10/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SM00002		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The long-term objective of the Medical Operations Support Team (MOST) is to develop support systems and processes consistent with Goal 9 in NASA's strategic plan—enable humans to live and work safely and effectively in space and to ensure crew health during space flight. The MOST specific aims are to: 1) maintain and expand the MOST team; 2) adapt the Human Patient Simulator physiologic scripting and modeling to reflect health threats likely to be encountered in space travel; 3) develop, evaluate, and validate medical procedures for appropriately responding to likely health threats; 4) develop an integrated, multimedia medical curriculum for preparing space- and ground-based personnel to respond to likely health threats; 5) secure peer review of MOST results; 6) support educational outreach activities; and 7) transfer technologies to the private sector. Moving beyond Phase 1 activities primarily focusing on specific aims 1-3, Phase 2 activities will focus on converting training scenarios developed around medical procedures for responding to selected health threats to instructional curricula that incorporate relevant and effective adult learning concepts and practices. Phase 2 will also be used to develop six new training scenarios dealing with likely health threats (e.g., cardiac		

Task Description:	events, acute abdominal events, decompression illnesses). The selected topics will be developed using a step-wise process refined in Phase 1: 1) topic identification; 2) learning objective formulation; 3) assessment metric generation; 4) scenario scripting and review; 4) scenario revision and script finalization; 5) instructional scenario filming/taping; and 6) prototype module assembly and preliminary testing. After testing in both ground-based analog situations, and when feasible, in parabolic flight tests on the KC-135, the scenarios will undergo expert review and refinement prior to being readied for conversion to professional learning curricula. Concurrent with developmental activities, active efforts around education and outreach to both adult and youth audiences will be underway, including readying manuscripts for submission to relevant scientific journals. Technology transfer to potential private sector users will also be stepped up, albeit in a cautious fashion to ensure integrity in the products being transferred, as well as in the transfer process. By the end of Phase 2, key components of an integrated instructional curriculum will be in place.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research in astronaut and flight surgeon training has obvious applications to terrestrial medical training. The same philosophy of operationally-relevant simulation-based training can be applied to the education programs of both traditional and non-traditional health care providers. In particular, our research into sustainment training will establish the standard for this type of testing. In addition, techniques developed by this team can be adapted by terrestrial health care providers; for example, adaptation of the KC-135 based free-floating incubation technique has been evaluated by paramedics as an alternative method for difficult airways. Modeling results are currently being used to create additional simulator-based patients for use in training anesthesiology students. We are also looking at Telementoring protocols in Resident education.
Task Progress:	The work completed to date in the MOST project reflects significant progress toward development of an integrated, web and simulation-based training program to prepare flight and ground-based personnel to cope effectively with anticipated health-related emergencies. Reviewed physiological research data, scripting and modeling literature. Scripted HPS physiological models for astronaut subpopulations. Developed, evaluated and validated, in a combination of ground-based analog test environments and during parabolic flight, several procedures relevant to space medicine using the enhanced, space adapted HPS. Demonstrated the utility of flight-ready HPS Gathered preliminary intubating laryngeal mask airway (ILMA) insertion data in both restrained and free-floating patients. Tested methods for unstowing the on-board defibrillator Evaluated non-clinicians' ability to conduct an in-flight papillary assessment. Evaluated the setup of intravenous fluid infusion system. Implemented and assessed the efficacy of the space-adapted HPS system as a training aid for knowledge and skill acquisition of procedures. Validated skill set training for surgical and non-surgical procedures (e.g., airway management as well as pneumothorax) using the HPS model. Developed training regimens for specific medical procedures (e.g., administering anesthesia to patients in the field) in response to requests from NASA personnel. Developed Flight Surgeon medical education,evaluation and training to provide core competancy training to them. Have also developed both ground and C-(training tor telementoring and telemedicine techniques. We have now started the ground based training and testing for CMO and astronaut physicians in telementoring environments. Project Number: SM00301 Principal Investigator: Harold K. Doerr, M.D.
Bibliography Type:	Description: (Last Updated: 10/22/2008)
Awards	N/A "invited as Key note at Wake Forest annual conference" Jan-2004
Awards	N/A "invited as the Keynote at the opening of the Riverside Simulation center along with the Govener of Ohio" Jan-2004
Awards	N/A "invited to give the plenary lecture at the National youth leadership forum" Jan-2004
Awards	N/A "invited to the ACGME Millennium conference as one of the countries top simulation experts" Jan-2004
Awards	N/A "RSI Anesthesia protocol accepted as operational by NASA and DDMS/DOD" Jan-2004
Presentation	Doerr, H "Can Anesthesia be Safely administered to Astronauts post flight" N/A Jun-2005
Presentation	Doerr, H "Considerations for a Modified rapid sequence induction in regards to Astronaut safetydurring a Post landing contingency." N/A Oct-2005
Presentation	Doerr, H "Medical simulations and Space Medicine: Engineering and Technology. Mars Mission era applied to Medical Training on Demand" N/A Feb-2005
Presentation	Doerr, H "Programming the simulator to a Space altered physiology" N/A Feb-2005
Presentation	Doerr, H "Simulation efforts at NASA" N/A Jun-2005
Presentation	Doerr, H "Simulation Models in the academic arena" N/A Aug-2005
Presentation	Doerr, H "Simulation seminar - Crisis resource management" N/A Sep-2005
Presentation	Doerr, H , Bacal, K "Mars is Closer than you think" N/A Nov-2005

Presentation	Doerr, H , Hurst, V "To Mars and beyond: You are the future of NASA." N/A Jul-2005
Presentation	Doerr, H , Stepaniac, P "Rapid sequence induction Anesthesia protocol" N/A Sep-2005